

## SAFETY DATA SHEET

Creation Date 06-July-2010

Revision Date 18-January-2018

Revision Number 3

### 1. Identification

**Product Name** SurePrep™ Soil DNA Purification Kit, Wash Solution I

**Cat No. :** BP2815-4; BP2815-50, BP2815-4E; BP2815-50E

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Manufacturer**

Fisher Scientific  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

##### **Emergency Telephone Number**

CHEMTREC®, Inside the USA:

800-424-9300

CHEMTREC®, Outside the USA:

001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                |              |
|------------------------------------------------|--------------|
| <b>Acute oral toxicity</b>                     | Category 4   |
| <b>Acute dermal toxicity</b>                   | Category 4   |
| <b>Acute Inhalation Toxicity</b>               | Category 4   |
| <b>Skin Corrosion/Irritation</b>               | Category 1 C |
| <b>Serious Eye Damage/Eye Irritation</b>       | Category 1   |
| <b>Health Hazards Not Otherwise Classified</b> | Category 1   |
| Contact with acids liberates very toxic gas    |              |
| Corrosive to the respiratory tract             |              |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled  
Causes severe skin burns and eye damage  
Contact with acids liberates very toxic gas  
Corrosive to the respiratory tract

**Precautionary Statements****Prevention**

Take any precaution to avoid mixing with acids  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wear respiratory protection  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER/doctor  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Harmful to aquatic life with long lasting effects  
Light sensitive

### 3. Composition/Information on Ingredients

| Component                                      | CAS-No    | Weight % |
|------------------------------------------------|-----------|----------|
| Water                                          | 7732-18-5 | 50-60    |
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0  | 40-50    |
| Sodium citrate                                 | 68-04-2   | 1 - 2    |
| Sarcocyl                                       | NA        | <1       |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash

|                                        |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | contaminated clothing before re-use. Call a physician immediately.                                                                                                                                                                                                                                                             |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                           |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | No information available                                          |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | No information available                                          |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Hydrogen cyanide (hydrocyanic acid) Nitrogen oxides (NO<sub>x</sub>) Sulfur oxides

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 4             | 1                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                               |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not flush into surface water or sanitary sewer system.                                                        |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.                                                                      |

## 7. Handling and storage

|                 |                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe vapors or spray mist. Do not ingest. |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage**

Corrosives area. Keep at temperatures between 15 ° and 30 °C. Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing before re-use. Wash hands before breaks and at the end of workday.

## 9. Physical and chemical properties

|                            |                          |
|----------------------------|--------------------------|
| <b>Physical State</b>      | Liquid                   |
| <b>Appearance</b>          | No information available |
| <b>Odor</b>                | No information available |
| <b>Odor Threshold</b>      | No information available |
| <b>pH</b>                  | 6.5-8.5                  |
| <b>Melting Point/Range</b> | No data available        |
| <b>Boiling Point/Range</b> | No information available |
| <b>Flash Point</b>         | No information available |
| <b>Evaporation Rate</b>    | No information available |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                                                           |
|----------------------------------|---------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                       |
| Stability                        | Stable under normal conditions. Light sensitive.                          |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to light.                    |
| Incompatible Materials           | Strong oxidizing agents, Acids                                            |
| Hazardous Decomposition Products | Hydrogen cyanide (hydrocyanic acid), Nitrogen oxides (NOx), Sulfur oxides |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                  |
| Hazardous Reactions              | None under normal processing.                                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                      |
|-------------|--------------------------------------|
| Oral LD50   | Category 4. ATE = 300 - 2000 mg/kg.  |
| Dermal LD50 | Category 4. ATE = 1000 - 2000 mg/kg. |
| Vapor LC50  | Category 4. ATE = 10 - 20 mg/l.      |

#### Component Information

| Component                                      | LD50 Oral                          | LD50 Dermal | LC50 Inhalation |
|------------------------------------------------|------------------------------------|-------------|-----------------|
| Water                                          | -                                  | Not listed  | Not listed      |
| Thiocyanic acid, compound with guanidine (1:1) | 282 mg/kg (rat)<br>593 mg/kg (rat) | Not listed  | Not listed      |
| Sodium citrate                                 | 5400 mg/kg (Mouse)                 | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                      | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------------------------------|-----------|------------|------------|------------|------------|------------|
| Water                                          | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Sodium citrate                                 | 68-04-2   | Not listed | Not listed | Not listed | Not listed | Not listed |
| Sarcocyl                                       | NA        | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                                                                                                  |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                  |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                  |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                                                                                                 |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                 |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                                      | Freshwater Algae                                      | Freshwater Fish                                        | Microtox                  | Water Flea                                      |
|------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|---------------------------|-------------------------------------------------|
| Thiocyanic acid, compound with guanidine (1:1) | Not listed                                            | Poecilia reticulata:<br>LC50=89.1 mg/L 96h             | Not listed                | EC50=42.4 mg/L 48h                              |
| Sodium citrate                                 | EC50: 18000 - 32000 mg/L,<br>96h (Chlorella vulgaris) | LC50: 18000 - 32000 mg/L,<br>96h (Poecilia reticulata) | EC50 1800 - 3200 mg/L 8 h | EC50: 5600 - 10000 mg/L,<br>48h (Daphnia magna) |

**Persistence and Degradability** Miscible with water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component                                      | log Pow |
|------------------------------------------------|---------|
| Thiocyanic acid, compound with guanidine (1:1) | -1.38   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                       |                          |
|-----------------------|--------------------------|
| UN-No                 | UN1760                   |
| Proper Shipping Name  | Corrosive liquid, n.o.s. |
| Proper technical name | (GUANIDINE THIOCYANATE)  |
| Hazard Class          | 8                        |
| Packing Group         | III                      |

### TDG

|                      |                         |
|----------------------|-------------------------|
| UN-No                | UN1760                  |
| Proper Shipping Name | Corrosive liquid, n.o.s |
| Hazard Class         | 8                       |
| Packing Group        | III                     |

### IATA

|       |        |
|-------|--------|
| UN-No | UN1760 |
|-------|--------|

|                      |                         |
|----------------------|-------------------------|
| Proper Shipping Name | Corrosive liquid, n.o.s |
| Hazard Class         | 8                       |
| Packing Group        | III                     |
| <b>IMDG/IMO</b>      |                         |
| UN-No                | UN1760                  |
| Proper Shipping Name | Corrosive liquid, n.o.s |
| Hazard Class         | 8                       |
| Packing Group        | III                     |

## 15. Regulatory information

**All of the components in the product are on the following Inventory lists:** No information available X = listed U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (ECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

### International Inventories

| Component                                      | DSL | NDSL | TSCA | EINECS    | ELINCS | PICCS | ENCS | AICS | KECL     | IECSC |
|------------------------------------------------|-----|------|------|-----------|--------|-------|------|------|----------|-------|
| Water                                          | X   | -    | X    | 231-791-2 | -      | X     | X    | X    | KE-35400 | X     |
| Thiocyanic acid, compound with guanidine (1:1) | X   | -    | X    | 209-812-1 | -      | X     | X    | X    | -        | X     |
| Sodium citrate                                 | X   | -    | X    | 200-675-3 | -      | X     | X    | X    | KE-20843 | X     |

### Legend

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## 16. Other information

**Prepared By** Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

**Creation Date** 06-July-2010  
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**Print Date** 18-January-2018

**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**